

02_ray

October 4, 2025

First: - go to the terminal - go to the `4_tuning_networks` folder, use `cd` to get there, or `z` to get there faster. - if you are planning on closing the laptop to let this run by itself, start a `tmux` session by giving the command `tmux`. You can name the session with `ctrl+b $`, leave the `tmux` with `ctrl+b d`, access it again with `tmux a -t 0` or replace the `0` with the name you gave it. See [tmux](#) for more info. - Activate the `.venv`, and run `python hypertune.py` - In the `hypertune.py` file, a `tune_dir` is specified: `models/ray`. We will check the contents of that folder after the `hypertune` finished. You can also use `tensorboard` to check the results. - this will take some time. I clocked 1901 seconds (30 minutes) on the VM

0.0992908 | 119 | 3 | 26 | 576.744 | 0 | 0.942187

```
[3]: from pathlib import Path
     tune_dir = Path("logs/ray").resolve()
     tune_dir.exists()
```

[3]: True

```
[4]: tunelogs = [d for d in tune_dir.iterdir()]
     tunelogs.sort()
     latest = tunelogs[-1]
     latest
```

[4]: PosixPath('/Users/nickreinders/Documents/Master_Applied_Data_Science_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-course/notebooks/4_tuning_networks/logs/ray/train_2025-10-04_08-55-56')

```
[5]: from ray.tune import ExperimentAnalysis
     import ray
     ray.init(ignore_reinit_error=True)

     analysis = ExperimentAnalysis(latest)
```

2025-10-04 16:39:59,692 INFO worker.py:1888 -- Started a local Ray instance.

So, we find some info:

```
[6]: analysis.results_df.columns
```

```
[6]: Index(['iterations', 'train_loss', 'test_loss', 'Accuracy', 'timestamp',  
        'checkpoint_dir_name', 'done', 'training_iteration', 'date',  
        'time_this_iter_s', 'time_total_s', 'pid', 'hostname', 'node_ip',  
        'time_since_restore', 'iterations_since_restore', 'experiment_tag',  
        'config/input_size', 'config/output_size', 'config/hidden_size',  
        'config/dropout', 'config/num_layers'],  
        dtype='object')
```

Let's focus on the parameters we wanted to tune.

```
[7]: import plotly.express as px  
  
plot = analysis.results_df  
select = ["Accuracy", "config/hidden_size", "config/dropout", "config/  
        ↪num_layers"]  
p = plot[select].reset_index().dropna()
```

Let's sort by accuracy

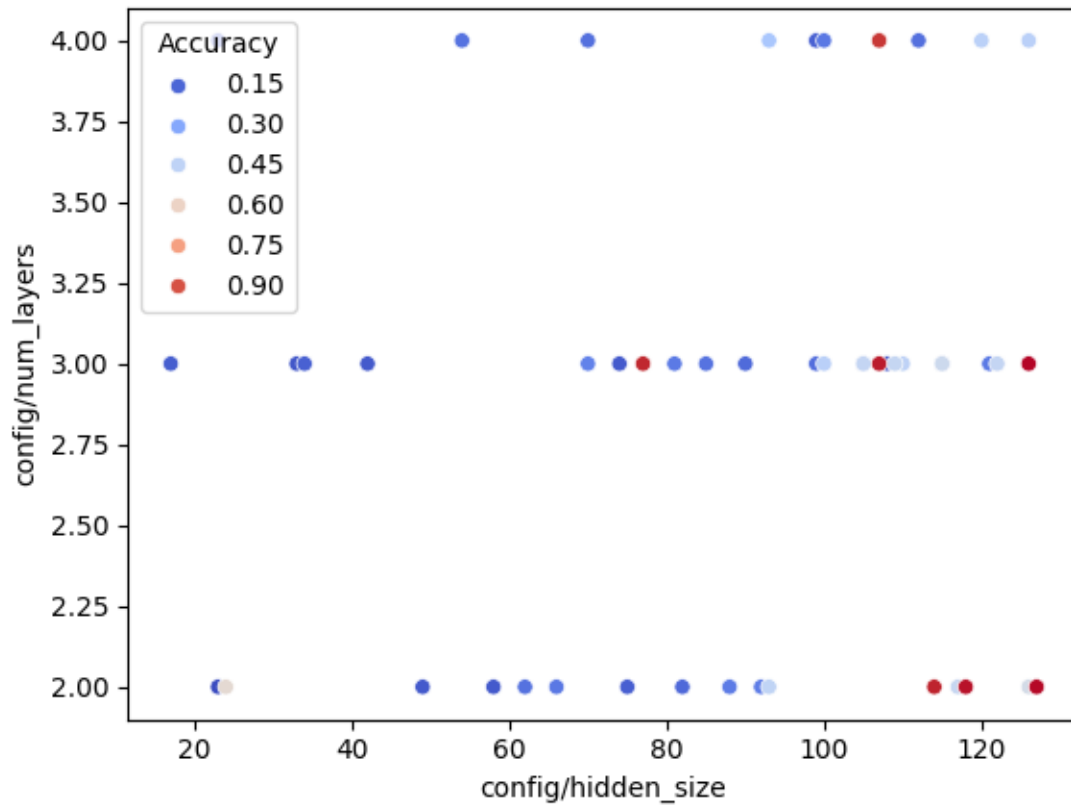
```
[8]: p.sort_values("Accuracy", inplace=True)
```

Make a parallel plot

```
[9]: px.parallel_coordinates(p, color="Accuracy")
```

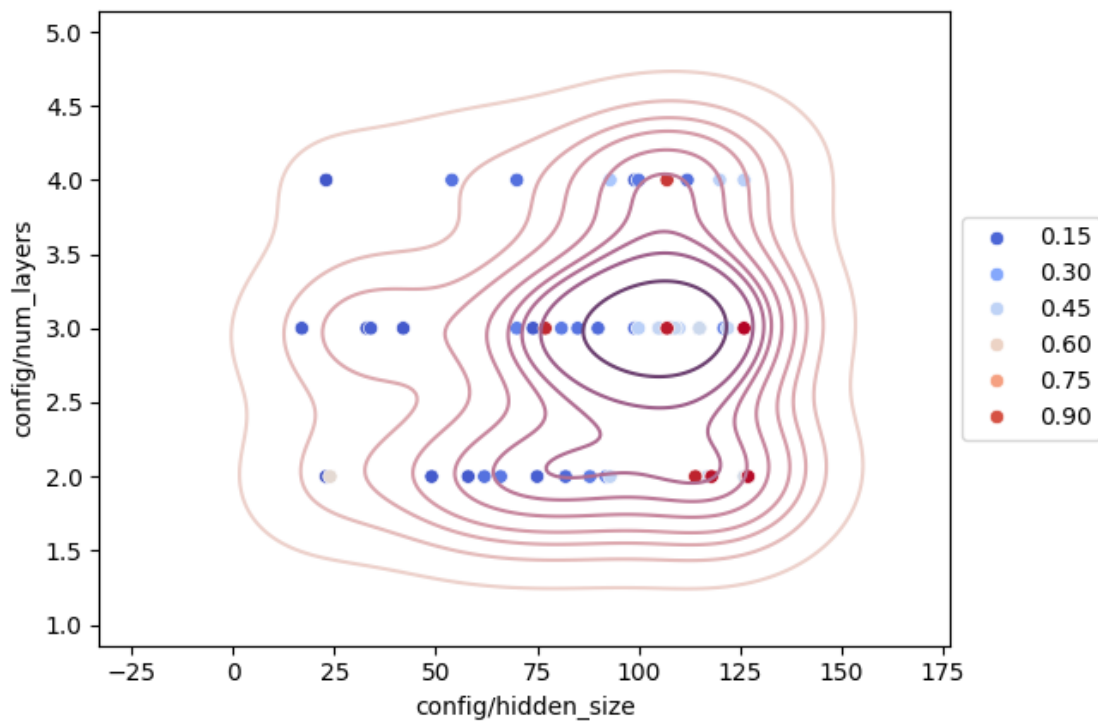
```
[10]: import seaborn as sns  
  
sns.scatterplot(data=p, x="config/hidden_size", y="config/num_layers",  
        ↪hue="Accuracy", palette="coolwarm")
```

```
[10]: <Axes: xlabel='config/hidden_size', ylabel='config/num_layers'>
```



```
[11]: import matplotlib.pyplot as plt
cmap = sns.cubehelix_palette(as_cmap=True)
sns.scatterplot(data=p, x="config/hidden_size", y="config/num_layers",
               hue="Accuracy", palette="coolwarm")
sns.kdeplot(data=p, x="config/hidden_size", y="config/num_layers", cmap=cmap)
plt.legend(loc='center left', bbox_to_anchor=(1, 0.5), ncol=1)
```

```
[11]: <matplotlib.legend.Legend at 0x125cc4f10>
```



Get the best trial

```
[12]: analysis.get_best_trial(metric="test_loss", mode="min")
```

```
[12]: train_cd3e9fd2
```

The top ten

```
[13]: p[-10:]
```

```
[13]:
```

	trial_id	Accuracy	config/hidden_size	config/dropout	config/num_layers
47	3121ca43	0.509375	126	0.090741	2
0	d54e7203	0.568750	24	0.097402	2
7	328e888e	0.937500	107	0.086803	4
5	4860f627	0.959375	77	0.026937	3
49	edf84a8e	0.964063	114	0.097899	2
1	3c8140af	0.970313	107	0.231316	3
48	9b93a537	0.978125	127	0.114222	2
8	690f8fe1	0.978125	118	0.158315	2
27	b776dd7d	0.982812	127	0.104135	2
31	cd3e9fd2	0.989062	126	0.293361	3

Or the best config

```
[14]: analysis.get_best_config(metric="Accuracy", mode="max")
```

```
[14]: {'input_size': 3,  
      'output_size': 20,  
      'data_dir': PosixPath('/Users/nickreinders/Documents/Master_Applied_Data_Scienc  
e_deeltijd_HAN/Deployment/Raoul_repo/MADS-MachineLearning-  
course/notebooks/4_tuning_networks/data/raw/gestures/gestures-dataset'),  
      'hidden_size': 126,  
      'dropout': 0.29336123692218813,  
      'num_layers': 3}
```